

LAB No 16

System Monitoring and Performance Tuning

Objectives

By the end of this lab, you will be able to:

1. Monitor system performance in real time using Linux commands.
2. Observe CPU, memory, and process utilization.
3. Interpret system statistics to identify performance bottlenecks.

Part 1: CPU Monitoring

◆ Command 1: `top`

Purpose: Shows a live view of CPU, memory, and process activity.

Steps:

1. Open the terminal.
2. Type:

```
bash
```

```
top
```

3. Observe:
 - The **first few lines** show CPU, memory, and load average.
 - The **lower part** lists running processes.

Expected Output

```
top - 14:35:20 up 1:20,  2 users,  load average: 0.10, 0.15, 0.12
Tasks: 210 total,  1 running, 209 sleeping,  0 stopped,  0 zombie
%Cpu(s): 2.0 us, 1.0 sy, 0.0 ni, 96.5 id, 0.3 wa
MiB Mem : 7950.0 total, 2000.0 free, 3500.0 used, 2450.0 buff/cache
```

Explanation:

- **us** = User processes
- **sy** = System (kernel) processes
- **id** = Idle percentage
- **wa** = Time waiting for disk/network

◆ Command 2: `top -b -n 1 > top_output.txt`

Purpose: Save a snapshot of system processes to a file.

Steps:

```
bash

top -b -n 1 > top_output.txt
cat top_output.txt
```

✓ *This creates a report you can view or submit.*

◆ Command 3: Filter for a Specific User

```
bash

top -u yourusername
```

Shows only the processes owned by that user.

◆ Command 4: Sort by Memory or CPU

While inside `top`:

- Press **M** → sort by memory
- Press **P** → sort by CPU

◆ Practice Summary

Task	Command	Purpose
View live usage	<code>top</code>	Real-time system stats
Save snapshot	<code>top -b -n 1 > top_output.txt</code>	Log processes
Filter user	<code>top -u user</code>	Show user-specific processes
Sort by memory	Press <code>M</code>	Find memory-heavy tasks
Kill process	Press <code>k</code>	Stop malfunctioning process

⚙️ Part 2: Enhanced CPU View using `htop`

◆ Installation

```
bash

sudo apt install htop  # Ubuntu/Debian
```

◆ Run htop

```
bash

htop
```

What You'll See:

- Colored bars for CPU, RAM, and Swap usage.
- Process list with PID, USER, CPU%, MEM%.

Key Actions

Key	Function
F6	Sort by CPU/MEM
F5	Tree View
F9	Kill Process
F10	Quit
F3	Search
F4	Filter



Part 3: CPU Statistics using mpstat



Command 1:

```
bash
```

```
mpstat
```

Shows CPU usage summary.

◆ Command 2:

```
bash  
  
mpstat 1 5
```

Displays CPU statistics every 1 second for 5 times.

Output Example:

```
perl  
  
10:45:23 AM  all   %usr %sys %iowait %idle  
10:45:23 AM  all   10.5  3.2   0.4   85.9
```

⚙️ Part 4: Memory Monitoring

◆ Command 1: `free -h`

Shows memory usage in a human-readable format.

```
bash  
  
free -h
```

Output Example:

```
vbnet  
  
              total    used    free   shared  buff/cache   available  
Mem:          15G       9.8G    2.0G    100M       3.2G         5.0G  
Swap:          4G       1.2G    2.8G
```



Interpretation:

- **used** = memory used by programs
- **buff/cache** = temporary kernel memory
- **available** = memory free for new programs

◆ Command 2: Continuous Monitoring

```
bash
```

```
free -s 5
```

Updates every 5 seconds.

◆ Command 3: Display in Megabytes

```
bash
```

```
free -m
```

◆ Command 4: Show Total RAM + Swap

```
bash
```

```
free -t
```

⚙️ Part 5: Virtual Memory Stats (vmstat)

Command 1:

```
bash

vmstat
```

Displays one-time statistics for memory, I/O, CPU, and swap.

Command 2:

```
bash

vmstat 2 5
```

Updates every 2 seconds × 5 times.



Sample Output:

CSS

```
procs -----memory----- --swap-- ----io----- -system-- -----cpu-----
r  b  swpd  free  buff  cache  si  so  bi  bo  in  cs  us  sy  id  wa  st
1  0      0 123456 34567 456789  0  0  12  4  78 162  2  1 97  0  0
```

Output Result:

abduallah-sohail@abduallah-sohail-VMware-Virtual-Platform: ~

abduallah-sohail@abduallah-sohail-VMware-Virtual-Platform:~\$ top

top - 10:18:19 up 1 min, 1 user, load average: 3.84, 1.61, 0.59

Tasks: 399 total, 1 running, 398 sleeping, 0 stopped, 0 zombie

%Cpu(s): 1.3 us, 0.6 sy, 0.0 ni, 98.1 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st

MiB Mem : 2852.7 total, 347.6 free, 1221.0 used, 1489.8 buff/cache

MiB Swap: 2852.0 total, 2851.7 free, 0.3 used, 1631.6 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5964	abdulla+	20	0	4195952	271296	132844	S	7.6	9.3	0:11.08	gnome-shell
7672	abdulla+	20	0	701508	60000	48300	S	1.3	2.1	0:01.44	gnome-terminal-
50	root	20	0	0	0	0	I	1.0	0.0	0:00.54	kworker/u513:2-flush-8:0
15	root	20	0	0	0	0	I	0.3	0.0	0:01.24	rcu_preempt
121	root	20	0	0	0	0	I	0.3	0.0	0:00.37	kworker/0:2-events
5755	abdulla+	9	-11	116024	14848	9176	S	0.3	0.5	0:00.52	pipewire
6222	abdulla+	20	0	214292	37968	30132	S	0.3	1.3	0:00.41	vntoolsd
7719	abdulla+	20	0	14524	6020	3764	R	0.3	0.2	0:00.08	top
1	root	20	0	23768	15040	9736	S	0.0	0.5	0:05.36	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.06	kthread
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0-cgroup_free
10	root	20	0	0	0	0	I	0.0	0.0	0:00.04	kworker/0:1-cgroup_free
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u512:0-ipv6_addrconf
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.13	ksoftirqd/0
15	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_gp

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top -b -n 1 > top_output.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat top_output.txt
top - 10:47:54 up 30 min, 1 user, load average: 0.17, 0.12, 0.17
Tasks: 343 total, 1 running, 342 sleeping, 0 stopped, 0 zombie
%Cpu(s): 4.1 us, 4.1 sy, 0.0 ni, 89.8 id, 0.0 wa, 0.0 hi, 2.0 si, 0.0 st
MiB Mem : 2852.7 total, 282.4 free, 1305.5 used, 1480.8 buff/cache
MiB Swap: 2852.0 total, 2851.7 free, 0.3 used, 1547.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
10985	abdulla+	20	0	14508	5656	3508	R	18.2	0.2	0:00.04	top
1	root	20	0	23768	15040	9736	S	0.0	0.5	0:06.08	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.07	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wor
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker+
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
14	root	20	0	0	0	0	S	0.0	0.0	0:00.15	ksoftir+
15	root	20	0	0	0	0	I	0.0	0.0	0:02.25	rcu_pre+
16	root	20	0	0	0	0	S	0.0	0.0	0:00.08	rcu_exp+
17	root	20	0	0	0	0	S	0.0	0.0	0:00.04	rcu_exp+
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.06	migrati+
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_in+
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
22	root	51	0	0	0	0	S	0.0	0.0	0:00.00	idle_in+

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top -u abdullah-sohail
```

```
top - 10:49:10 up 32 min, 1 user, load average: 0.34, 0.15, 0.18
Tasks: 345 total, 1 running, 344 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.7 us, 1.3 sy, 0.0 ni, 96.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 2852.7 total, 285.7 free, 1301.5 used, 1481.5 buff/cache
MiB Swap: 2852.0 total, 2851.7 free, 0.3 used, 1551.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5964	abdulla+	20	0	4226792	294988	132628	S	8.2	10.1	1:00.83	gnome-shell
7672	abdulla+	20	0	703992	62008	48924	S	5.9	2.1	0:15.90	gnome-terminal
11050	abdulla+	20	0	14528	6004	3764	R	2.3	0.2	0:00.82	top
11085	abdulla+	39	19	474340	31036	25096	S	0.3	1.1	0:00.20	tracker-extract
5739	abdulla+	20	0	21300	12856	9736	S	0.0	0.4	0:03.95	systemd
5742	abdulla+	20	0	21476	3656	1848	S	0.0	0.1	0:00.00	(sd-pam)
5755	abdulla+	9	-11	116024	14692	9020	S	0.0	0.5	0:00.53	pipewire
5757	abdulla+	20	0	97744	6176	5292	S	0.0	0.2	0:00.03	pipewire
5759	abdulla+	20	0	47748	14260	12548	S	0.0	0.5	0:02.29	snapd-desktop-i
5763	abdulla+	9	-11	406828	19004	13996	S	0.0	0.7	0:00.33	wireplumber
5771	abdulla+	9	-11	116108	13128	9500	S	0.0	0.4	0:00.14	pipewire-pulse
5772	abdulla+	20	0	11036	6904	4776	S	0.0	0.2	0:02.65	dbus-daemon
5773	abdulla+	20	0	316516	10596	9496	S	0.0	0.4	0:00.15	gnome-keyring-d
5814	abdulla+	20	0	536568	7696	6912	S	0.0	0.3	0:00.13	xdg-document-po
5818	abdulla+	20	0	309316	6316	5792	S	0.0	0.2	0:00.03	xdg-permission-
5825	abdulla+	20	0	235676	6240	5736	S	0.0	0.2	0:00.01	gdm-wayland-ses
5834	abdulla+	20	0	298244	16876	14940	S	0.0	0.6	0:00.07	gnome-session-b
5901	abdulla+	20	0	162660	6996	6300	S	0.0	0.2	0:00.01	gcr-ssh-agent
5902	abdulla+	20	0	91556	5780	5324	S	0.0	0.2	0:00.01	gnome-session-c
5922	abdulla+	20	0	314300	8420	7444	S	0.0	0.3	0:00.12	gvfsd
5928	abdulla+	20	0	459716	7416	6612	S	0.0	0.3	0:00.02	gvfsd-fuse
5931	abdulla+	20	0	602440	18804	16060	S	0.0	0.6	0:00.45	gnome-session-b

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ http
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```



```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~

0[|
1[|
2[|
3[|
Mem[|
Swp[|

0.6%] Tasks: 131, 418 thr, 209 kthr; 1 running
0.0%] Load average: 0.11 0.09 0.13
1.8%] Uptime: 00:40:37
10.4%
1.09G/2.79G
0K/2.79G

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ CommandΔ
1 root 20 0 23768 15040 9736 S 0.0 0.5 0:06.23 /sbin/init splash
1234 root 20 0 1950M 41460 26376 S 0.0 1.4 0:04.47 /snap/snapd/current/usr/lib/snapd/snapd
1744 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.18 /snap/snapd/current/usr/lib/snapd/snapd
1747 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.72 /snap/snapd/current/usr/lib/snapd/snapd
1748 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
1749 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.43 /snap/snapd/current/usr/lib/snapd/snapd
1751 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
1918 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.36 /snap/snapd/current/usr/lib/snapd/snapd
1963 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.72 /snap/snapd/current/usr/lib/snapd/snapd
2125 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.07 /snap/snapd/current/usr/lib/snapd/snapd
2609 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.51 /snap/snapd/current/usr/lib/snapd/snapd
4349 root 20 0 1950M 41460 26376 S 0.0 1.4 0:00.29 /snap/snapd/current/usr/lib/snapd/snapd
807 root 20 0 56076 12408 10840 S 0.0 0.4 0:00.11 /usr/bin/VGAAuthService
2652 pcpc 20 0 14632 7244 6176 S 0.0 0.2 0:00.10 /usr/bin/pmie -b -m pmie_check -F -P -l /var/log/pcp/pmie/abdullah-sohail-VMware-Virt
1213 root 20 0 40656 23680 12348 S 0.0 0.8 0:00.92 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
1762 root 20 0 115M 25376 15460 S 0.0 0.9 0:00.36 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-fo
1900 root 20 0 115M 25376 15460 S 0.0 0.9 0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait
819 root 20 0 239M 9548 8084 S 0.6 0.3 0:06.41 /usr/bin/vmtoolsd
906 root 20 0 239M 9548 8084 S 0.0 0.3 0:00.00 /usr/bin/vmtoolsd
916 root 20 0 239M 9548 8084 S 0.0 0.3 0:00.21 /usr/bin/vmtoolsd
F1Help F2Setup F3Search F4Filter F5List F6SortBy F7Nice F8Nice F9Kill F10Quit
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ htop
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mpstat
Linux 6.17.0-19-generic (abdullah-sohail-VMware-Virtual-Platform) 05/08/2026 _x86_64_ (4 CPU)

10:59:02 AM CPU %usr %nice %sys %iowait %irq %soft %steal %guest %gnice %idle
10:59:02 AM all 1.25 0.29 2.05 0.09 0.00 0.07 0.00 0.00 0.00 96.25
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mpstat 1 5
Linux 6.17.0-19-generic (abdullah-sohail-VMware-Virtual-Platform) 05/08/2026 _x86_64_ (4 CPU)

10:59:21 AM CPU %usr %nice %sys %iowait %irq %soft %steal %guest %gnice %idle
10:59:22 AM all 1.77 0.00 0.76 0.00 0.00 0.00 0.00 0.00 0.00 97.47
10:59:23 AM all 0.51 0.00 1.27 0.00 0.00 0.00 0.00 0.00 0.00 98.23
10:59:24 AM all 0.50 0.25 0.76 0.00 0.00 0.00 0.00 0.00 0.00 98.49
10:59:25 AM all 0.75 0.00 0.50 0.00 0.00 0.00 0.00 0.00 0.00 98.75
10:59:26 AM all 0.25 0.00 0.25 0.00 0.00 0.00 0.00 0.00 0.00 99.50
Average: all 0.75 0.05 0.70 0.00 0.00 0.00 0.00 0.00 0.00 98.49
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -h
total used free shared buff/cache available
Mem: 2.8Gi 1.3Gi 277Mi 40Mi 1.5Gi 1.5Gi
Swap: 2.8Gi 316Ki 2.8Gi

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -s 5
total used free shared buff/cache available
Mem: 2921116 1323880 286348 41784 1532120 1597236
Swap: 2920444 316 2920128

total used free shared buff/cache available
Mem: 2921116 1323984 286244 41784 1532128 1597132
Swap: 2920444 316 2920128

total used free shared buff/cache available
Mem: 2921116 1323984 286244 41784 1532128 1597132
Swap: 2920444 316 2920128

total used free shared buff/cache available
Mem: 2921116 1323976 286244 41784 1532128 1597140
Swap: 2920444 316 2920128

total used free shared buff/cache available
Mem: 2921116 1323976 286244 41784 1532128 1597140
Swap: 2920444 316 2920128

total used free shared buff/cache available
Mem: 2921116 1334524 275596 41784 1532232 1586592
```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -m
              total        used        free      shared  buff/cache   available
Mem:           2852         1285         286         40         1496         1567
Swap:          2851           0        2851
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ free -t
              total        used        free      shared  buff/cache   available
Mem:          2921116      1316152      293476       41784      1532756      1604964
Swap:          2920444         316      2920128
Total:         5841560      1316468      3213604
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st gu
 2 0   316 291596 32540 1500524 0 0 747 234 422 0 1 2 97 0 0 0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ vmstat 2 5
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st gu
 2 0   316 291288 32556 1500596 0 0 741 233 421 0 1 2 97 0 0 0
 0 0   316 291288 32556 1500596 0 0 0 0 320 320 1 1 99 0 0 0
 0 0   316 291288 32564 1500596 0 0 0 6 138 147 0 0 99 0 0 0
 0 0   316 291288 32564 1500596 0 0 0 0 547 588 1 1 98 0 0 0
 0 0   316 291288 32564 1500596 0 0 0 2 158 156 0 0 99 0 0 0
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```